



PIYUSH GIRISH KULKARNI

Address: Ölmühlenweg 33B, 22047, Hamburg, Germany.

Phone: +4915560762743

Email – kulpiyush15@gmail.com | LinkedIn: [LinkedIn](#) | Website: [Portfolio](#)

PROFILE SUMMARY

AI & Data enthusiast with hands-on experience engineering both rapid prototypes and production-grade GenAI solutions using state-of-the-art LLMs. Proven track record of bridging technology and business, translating complex AI capabilities into accessible tools, reliable workflows, and actionable client enablement.

SKILLS

- LLM & GenAI Ecosystem: OpenAI (GPT-4 / ChatGPT), Google Gemini, Prompt Engineering, RAG Architectures, LangChain, Agentic Systems, MCP Servers, REST APIs.
- AI Prototyping & Tools: Streamlit, Python (Pandas, NumPy), Power BI, Automation Workflows, Data Visualization.
- Databases & Infrastructure: SQL, PostgreSQL, ChromaDB (Vector DB), Git, AWS, Docker, Airflow, Dagster.

WORK EXPERIENCE

GenAI & Agentic AI Engineer (Company Project) | Galagos AI – Hamburg, Germany | *March 2025 – August 2025*

- Prototyping & Technical Leadership: Led a 5-member team to architect prototype-ready search engines using Google Gemini, LangChain, and ChromaDB, translating complex domain queries into automated workflow and tool recommendations.
- Prompt Engineering & Agent Optimization: Leveraged Gemini APIs within autonomous Self-RAG agent frameworks, designing targeted prompts and evaluation layers to iteratively improve context retrieval accuracy for specialized terminology.
- Technical Enablement: Directed architectural choices for automated data ingestion and platform integrations (via MCP servers and EXA Search), bridging technical implementation with user-centric functionality.

Data Scientist | Quantum Innotek Solutions – India | *Dec 2023 – Aug 2024*

- Algorithm Development: Engineered a targeted search and matching algorithm utilizing KNN and FLAN to process complex input data structures and deliver highly relevant results.
- Backend Intelligence: Built custom relational dictionary databases and data access layers via SQLAlchemy and MySQL, developing Flask backends to manage model operations.
- CI/CD Deployment: Enabled seamless continuous deployment of algorithmic models to Linode production servers by building and maintaining a customized Git-based CI/CD pipeline.

Software Engineer & Data Analyst | CIS IT Solutions Pvt Ltd – India | *Feb 2023 – Dec 2023*

- Predictive Modeling: Developed custom machine learning models using Python and NumPy to significantly enhance client predictive analytics and model forecasting accuracy.
- Data Exploration: Supported epidemiological research by conducting deep exploratory data analysis (EDA) and engineering interactive data visualizations using Power BI and MS Excel.
- Workflow Automation: Streamlined client operations and data collection capabilities by developing automated cloud workflow applications using Microsoft Power Apps and AWS.

Data Engineer (Working Student) | STATISTA GmbH – Hamburg, Germany | *July 2025 – Current*

- AI-Driven Pipeline Automation: Engineered automated ETL/ELT pipelines using Python and REST APIs, integrating OpenAI GPT models for automated metadata extraction, text classification, and dataset structuring across 20+ international economic sources.
 - Performance & Quality Optimization: Implemented prompt templates and chunked processing in Pandas to efficiently process millions of raw records while maintaining high data quality standards.
-

EDUCATION

SRH University of Heidelberg Campus Hamburg, Hamburg, Germany

Master of Science – Applied Data Science & Analytics (Oct 2024 – Present)

- Relevant Coursework: Machine Learning, Deep Learning, Big Data with Python, Statistics, Data Analysis, Data Management.

MIT ADT University, Pune, India

Bachelor of Technology – Computer Science & Engineering (Class of 2022)

- Relevant Coursework: Core AI Technologies, Deep Learning, Machine Learning.

ACADEMIC PROJECTS

FoodVisionAI-Food Recognition & Nutrition App - [Link](#)

- Vision Architecture: Fine-tuned PyTorch CNNs (EfficientNet) on a large-scale dataset of 131K+ images to achieve a 95% real-time dish classification accuracy.
- Semantic Matching Engine: Utilized Sentence Transformers to accurately map visual predictions to a 1,000+ item nutrition database with an ultra-low query latency of ~14ms.
- GenAI Deployment: Built a responsive Streamlit web application integrating local LLMs (Ollama) to deliver interactive AI-driven Q&A and comprehensive food descriptions.

Topic Modelling Using LDA (July 2021 to May 2022) - [Link](#)

- NLP Framework: Utilized Latent Dirichlet Allocation (LDA) frameworks to rapidly process, cluster, and analyze extensive corpuses of unstructured biomedical text data.
- Summarization Architecture: Built a specialized question-answering model leveraging the PICO framework for precise context extraction from dense COVID-19 datasets.
- Text Processing Pipelines: Engineered automated regex, text tokenization, and vectorization pipelines to transform raw literature into structured machine-readable formats.

IoT Smoke Detection Data Pipeline- [LINK](#)

- ML Integration: Developed and deployed automated Scikit-learn predictive models via REST APIs to handle real-time classification inference.
- Streaming Infrastructure: Built a live event-streaming and batch pipeline using Apache Kafka, Docker, and Airflow to stream high-frequency IoT sensor metrics.
- Model Observability: Integrated Prometheus and Grafana for live data drift monitoring and achieved 95% automated test coverage across the model environment.

ACHIEVEMENTS

- AWS Certified Cloud Practitioner
- Copyright: [Perspective Approach for Mango Ripening Classifiers](#)

LANGUAGES

- English – Native
- German – Conversational (A2 – B1)



Place: Hamburg, Germany

Date: 26-06-2026